

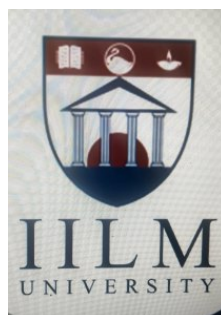
Report
Of
Internship

**DESIGN AND DEVELOPMENT OF THE BHARAT COMPUTER
INSTITUTE WEBSITE USING HTML, CSS, JAVASCRIPT, NODE.JS
AND MYSQL**

Submitted
To

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech Computer Science and Engineering



By

KARAN ARYA

Roll Number: 2CSC1003002587

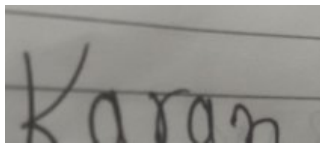
Session: 2026



CANDIDATE'S DECLARATION

I, KARAN ARYA, do hereby declare that the internship report titled "DESIGN AND DEVELOPMENT OF THE BHARAT COMPUTER INSTITUTE WEBSITE USING HTML, CSS, JAVASCRIPT, NODE.JS AND MYSQL" has been completed by me to fulfil the requirement for the award of my degree. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

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Student Name: KARAN ARYA

Father's Name: ADITYA KUMAR

Roll No.: 2CSC1003002587

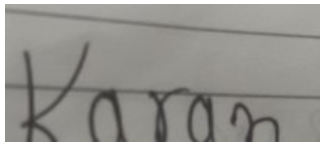
ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship programme. I am thankful for their guidance, constructive feedback, and encouragement during this work.

I would like to thank the training and mentoring team at Bharat Computer Institute, Rohtak Road, Jhajjar, Haryana, for the opportunity to intern in Web Development and for their support throughout the internship.

I also express my gratitude to my parents and teachers for their constant encouragement and blessings.

Signature:

A photograph of a handwritten signature in black ink on a light-colored background. The signature is written in a cursive style and appears to read 'Karan'.

Student Name: KARAN ARYA

Roll No.: 2CSC1003002587

INTERNSHIP COMPLETION CERTIFICATE

The certificate below was issued by the E-Max Council for Vocational Training and Education, confirming successful completion of the Web Development internship that forms the basis of this report.

Certificate Details:

Session Duration: 120 Hours

Course: Internship on Web Development

Institute: Bharat Computer Institute (Rohtak Road, Jhajjar, Haryana, EMAX/FP99799)

Subjects Covered: Front-End (Client-Side), Back-End (Server-Side), Core Tools & Concepts

Marks Secured: 90% with A+ Grade

Issue Date: 21 July 2026

Roll No. (Certificate): EMCE1040914 | Enrollment No.: 04-EC-405293

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Enrollment No. : 04-EC-405293

ई-मैक्स व्यावसायिक शिक्षा प्रशिक्षण परिषद्

E-Max Council for Vocational Training and Education

(An Autonomous Body Registered Under The SR Act Section 9(1) Act, 2012)
CR Act Department for Promotion of I.I.T. Govt. of India
Govt. Regn. No. HR/KKR/2015/00328





E-MAX INDIA

CERTIFICATE

Session: 120 Hours

This is to certify **KARAN ARYA**
 Father's Name **ADITYA KUMAR** Issue Date **21 July 2026**
 Has completed **Internship on Web Development**
 At **BHARAT COMPUTER INSTITUTE (Rohtak Road, Jhajjar, Haryana, EMAX/FP99799)**
 with the following subjects **Front-End (Client-Side), Back-End (Server-Side), Core Tools & Concepts.**
 He has secured **90% Marks**
 With **A+** Grade


Controller Of Exam


E-Max Education
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E-Max Council Certificate — Internship on Web Development, Karan Arya.

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report documents the Web Development internship undertaken at Bharat Computer Institute (Rohtak Road, Jhajjar, Haryana), covering a session of 120 hours, as part of the practical/internship component of my academic programme. The internship, certified by the E-Max Council for Vocational Training and Education, covered three core areas of web development: Front-End (Client-Side) development, Back-End (Server-Side) development, and Core Tools & Concepts.

As the practical deliverable for this internship, I designed and developed the official website for Bharat Computer Institute itself (live at bharat-computer-institute-jhajjar.netlify.app) — covering the Home, About Us, and Courses pages — using HTML, CSS, and JavaScript on the front end, Node.js for server-side logic, Python for supporting automation scripts, and MySQL as the backend database, with the finished site deployed live on Netlify.

4.2 Organization Profile

Bharat Computer Institute, located on Rohtak Road, Jhajjar, Haryana, is a training centre affiliated with the E-Max Council for Vocational Training and Education (an autonomous body registered under the SR Act, Section 9(1), 2012, Govt. Regn. No. HR/KKR/2015/00328). The institute conducts vocational and job-oriented training programmes, including a structured 120-hour Web Development internship covering front-end technologies, back-end/ server-side development, and the core tools and concepts used in modern web projects.

4.3 Problem Statement

Bharat Computer Institute needed a modern, mobile-responsive website to reach prospective students, present its certification programmes (Certificate, Diploma, and Advance Diploma courses), and build credibility as “Jhajjar's Most Trusted Computer Institute.” A well-structured, responsive website that clearly presents information, allows visitors to get in touch, and stores enquiries reliably addresses this gap. This internship project set out to design and build such a website end-to-end — from front-end pages through to a working backend that can store form submissions in a database — and to deploy it so it is publicly accessible.

4.4 Project Objectives

- To design and build a responsive, multi-page institute website front end using semantic HTML5 and CSS3, including a mobile-friendly layout.

- To add interactivity (navigation, form validation, dynamic content) using JavaScript.

- To build a Node.js backend that accepts contact/admission enquiry submissions and serves data via simple API routes.

- To design a MySQL database schema to store enquiries/user submissions, and to

connect it to the Node.js backend.

To use Python for small supporting/automation scripts (e.g., data formatting, basic testing of API responses).

To deploy the completed website to production using Netlify, and verify it functions correctly live.

4.5 Scope of the Project

The scope of this internship covers the design, development, and deployment of the Bharat Computer Institute website as a self-contained project. It includes the front-end pages (Home, About Us, Courses, Admission), a working backend for form handling, database integration, and live deployment. It does not include advanced features such as user authentication, payment integration, or a full content-management system, which were outside the time and scope of the 120-hour internship.

4.6 Technologies and Tools Used

Layer / Component	Technology / Tool Used
Front-End Structure	HTML5
Front-End Styling	CSS3 (Flexbox, Grid, Media Queries for responsiveness)
Front-End Interactivity	JavaScript (DOM manipulation, form validation, fetch API)
Back-End Runtime	Node.js with Express.js
Database	MySQL
Supporting Scripts	Python (data formatting / basic API testing scripts)
Deployment	Netlify
Development Environment	VS Code
Version Control	Git and GitHub

Table 4.1: Technologies and tools used in the project.

Module	Purpose
Home Page	Introduces the business/individual, highlights key services
About / Services Pages	Provides details of offerings using responsive HTML/CSS layouts
Contact Form	Validated with JavaScript, submits data to the Node.js backend
Node.js API	Receives form data, inserts it into the MySQL database
MySQL Database	Stores enquiries/submissions for later retrieval

Table 4.2: Website modules and their purpose.

4.7 System Architecture

The website follows a simple client-server architecture. The front end (HTML, CSS, JavaScript) runs in the browser and communicates with a Node.js/Express backend over

HTTP. The backend exposes an API route that receives contact-form submissions, validates them server-side, and stores them in a MySQL database. The completed application was deployed on Netlify so it is accessible over the internet.

Figure 4.0 illustrates the system architecture: the front end communicates with the Node.js/Express API, which reads from and writes to the MySQL database, and the finished application is deployed live on Netlify.

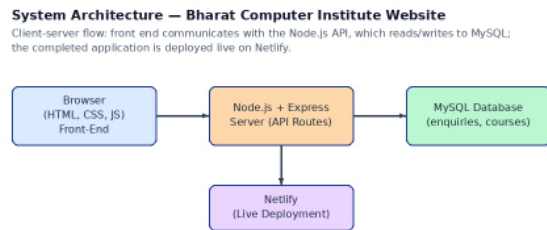


Figure 4.0: System architecture — Bharat Computer Institute website.

4.8 Methodology

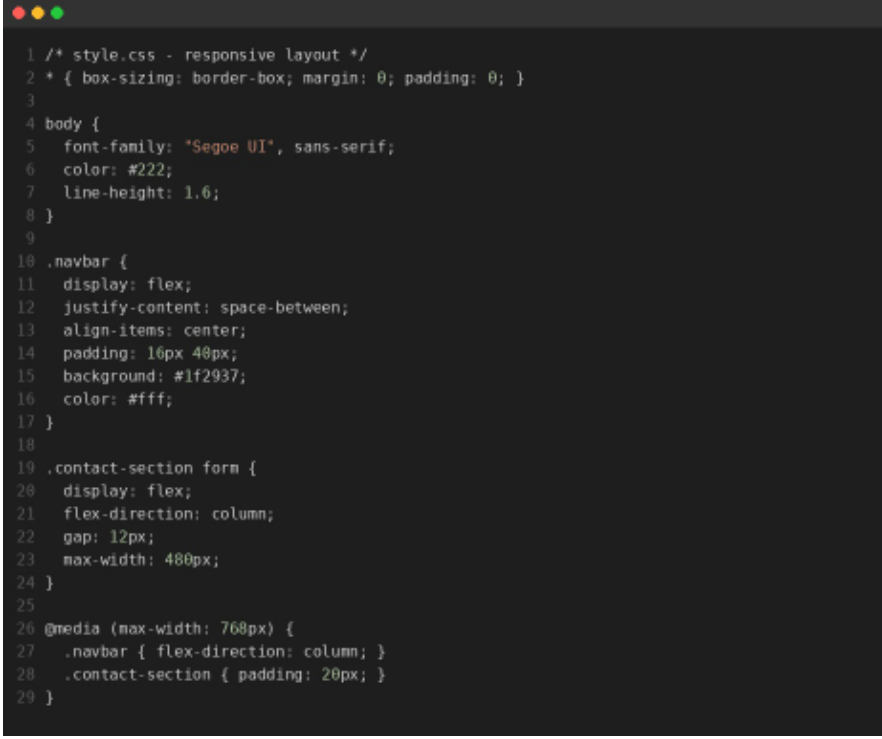
4.8.1 Front-End Development

The front end was built using semantic HTML5 for page structure, CSS3 (Flexbox and Grid) for a responsive layout that adapts to desktop and mobile screens, and JavaScript for interactive behaviour such as menu toggles, smooth scrolling, and client-side validation of the contact form before submission. Representative snippets are shown in Figures 4.1–4.3.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Karan Arya | Portfolio & Business Site</title>
7   <link rel="stylesheet" href="css/style.css">
8 </head>
9 <body>
10  <header class="navbar">
11    <h1 class="logo">MyBusiness</h1>
12    <nav>
13      <a href="#home">Home</a>
14      <a href="#services">Services</a>
15      <a href="#contact">Contact</a>
16    </nav>
17  </header>
18
19  <section id="contact" class="contact-section">
20    <h2>Get in Touch</h2>
21    <form id="contactForm">
22      <input type="text" id="name" placeholder="Your Name" required>
23      <input type="email" id="email" placeholder="Your Email" required>
24      <textarea id="message" placeholder="Message" required></textarea>
25      <button type="submit">Send Message</button>
26    </form>
27  </section>
28  <script src="js/main.js"></script>
29 </body>
30 </html>
  
```

Figure 4.1: index.html — page structure and contact form (sample).

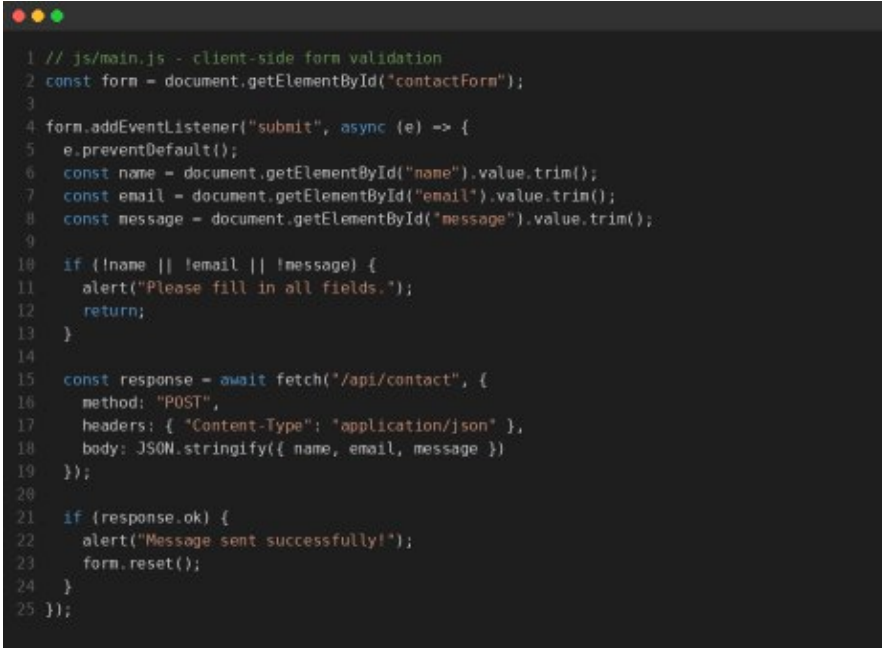


```

1 /* style.css - responsive layout */
2 * { box-sizing: border-box; margin: 0; padding: 0; }
3
4 body {
5   font-family: "Segoe UI", sans-serif;
6   color: #222;
7   line-height: 1.6;
8 }
9
10 .navbar {
11   display: flex;
12   justify-content: space-between;
13   align-items: center;
14   padding: 16px 40px;
15   background: #1f2937;
16   color: #fff;
17 }
18
19 .contact-section form {
20   display: flex;
21   flex-direction: column;
22   gap: 12px;
23   max-width: 480px;
24 }
25
26 @media (max-width: 768px) {
27   .navbar { flex-direction: column; }
28   .contact-section { padding: 20px; }
29 }

```

Figure 4.2: style.css — responsive layout rules (sample).



```

1 // js/main.js - client-side form validation
2 const form = document.getElementById("contactForm");
3
4 form.addEventListener("submit", async (e) => {
5   e.preventDefault();
6   const name = document.getElementById("name").value.trim();
7   const email = document.getElementById("email").value.trim();
8   const message = document.getElementById("message").value.trim();
9
10  if (!name || !email || !message) {
11    alert("Please fill in all fields.");
12    return;
13  }
14
15  const response = await fetch("/api/contact", {
16    method: "POST",
17    headers: { "Content-Type": "application/json" },
18    body: JSON.stringify({ name, email, message })
19  });
20
21  if (response.ok) {
22    alert("Message sent successfully!");
23    form.reset();
24  }
25 });

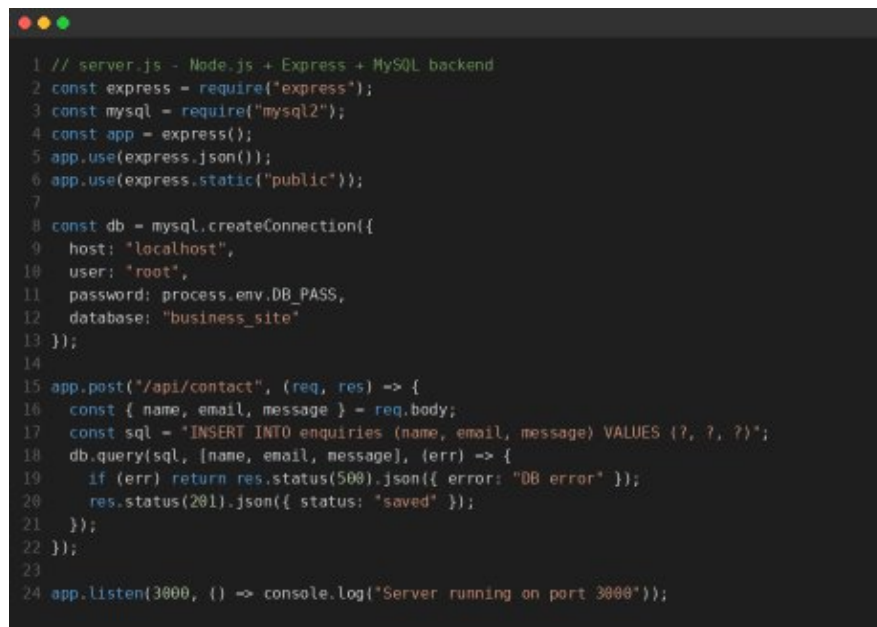
```

Figure 4.3: main.js — client-side form validation and fetch call (sample).

4.8.2 Back-End Development

A Node.js server using the Express.js framework was built to handle incoming requests. A POST route was created to accept contact-form data (name, email, message) from the front end, perform basic server-side validation, and insert the record into a MySQL table using a

Node.js MySQL client library. A corresponding GET route was used during development to confirm that records were being stored correctly. Figure 4.4 shows a representative version of this route.



```

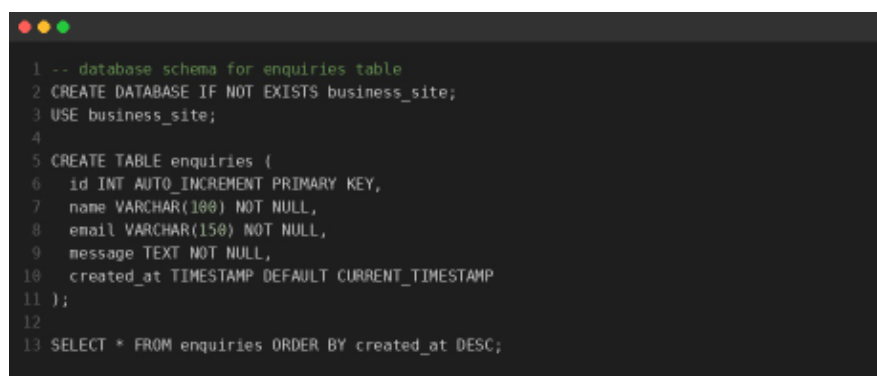
1 // server.js - Node.js + Express + MySQL backend
2 const express = require("express");
3 const mysql = require("mysql2");
4 const app = express();
5 app.use(express.json());
6 app.use(express.static("public"));
7
8 const db = mysql.createConnection({
9   host: "localhost",
10  user: "root",
11  password: process.env.DB_PASS,
12  database: "business_site"
13 });
14
15 app.post("/api/contact", (req, res) => {
16   const { name, email, message } = req.body;
17   const sql = `INSERT INTO enquiries (name, email, message) VALUES (?, ?, ?)`;
18   db.query(sql, [name, email, message], (err) => {
19     if (err) return res.status(500).json({ error: "DB error" });
20     res.status(201).json({ status: "saved" });
21   });
22 });
23
24 app.listen(3000, () => console.log("Server running on port 3000"));

```

Figure 4.4: server.js — Express route connecting to MySQL (sample).

4.8.3 Database Design

A MySQL database was created with a table to store contact-form submissions, including fields for name, email, message, and submission timestamp. Basic queries (INSERT, SELECT) were used to verify that data flowing from the front end through the Node.js API was being persisted correctly. Figure 4.5 shows the schema used.



```

1 -- database schema for enquiries table
2 CREATE DATABASE IF NOT EXISTS business_site;
3 USE business_site;
4
5 CREATE TABLE enquiries (
6   id INT AUTO_INCREMENT PRIMARY KEY,
7   name VARCHAR(100) NOT NULL,
8   email VARCHAR(150) NOT NULL,
9   message TEXT NOT NULL,
10  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
11 );
12
13 SELECT * FROM enquiries ORDER BY created_at DESC;

```

Figure 4.5: schema.sql — enquiries table definition (sample).

4.8.4 Python Scripts

Python was used for small supporting tasks outside the main application — such as formatting sample data used while testing the contact form, and writing simple scripts to send test requests to the API endpoints and check the responses during development.

4.8.5 Deployment

The completed project was pushed to a GitHub repository and deployed using Netlify at bharat-computer-institute-jhajjar.netlify.app. Environment variables (such as database connection details) were configured for the live deployment, and the site was tested to confirm that it renders and behaves correctly in production. Figures 4.6–4.10 show the live deployed website, page by page, along with a short description of what each page does.

Figure 4.6 — Homepage: This is the first page a visitor sees on opening the site. It shows the institute's hero banner with the tagline “Shaping the Future with Skills” and a “Start Your Journey Today” call-to-action button that leads to the Admission page. A green announcement strip runs across the top with the current batch start date and the institute's contact number, and a fixed navigation bar (Home, About Us, Courses, Admission, Contact Us) stays visible as the visitor scrolls.

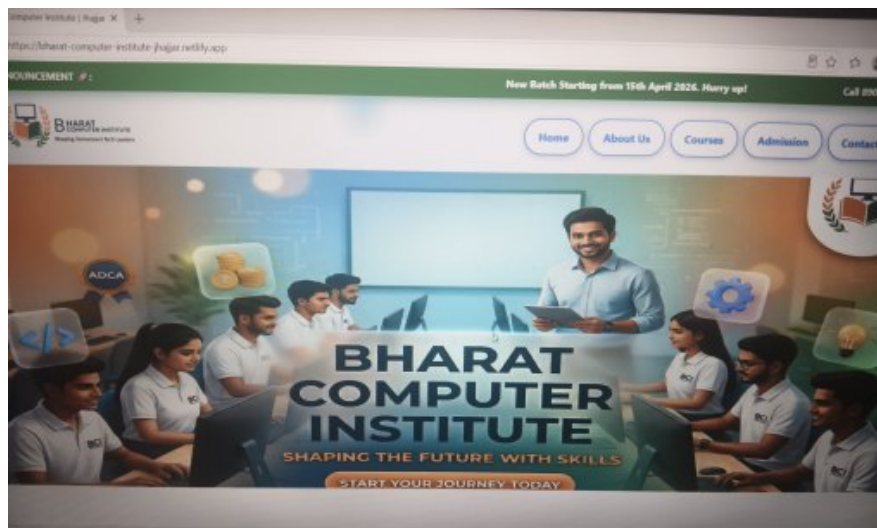


Figure 4.6: Live homepage — bharat-computer-institute-jhajjar.netlify.app.

Figure 4.7 — Courses page: Clicking “Courses” in the navigation bar opens this page. It lists the institute's three certification tracks — Certificate Courses, Diploma Courses, and Advance Diploma — each with a short description and a “Learn More” button that a visitor can click to see full course details. A stats strip above the cards highlights the institute's track record: 5,000+ students trained, 100+ courses, 100% placement assist, and 10+ years of experience.

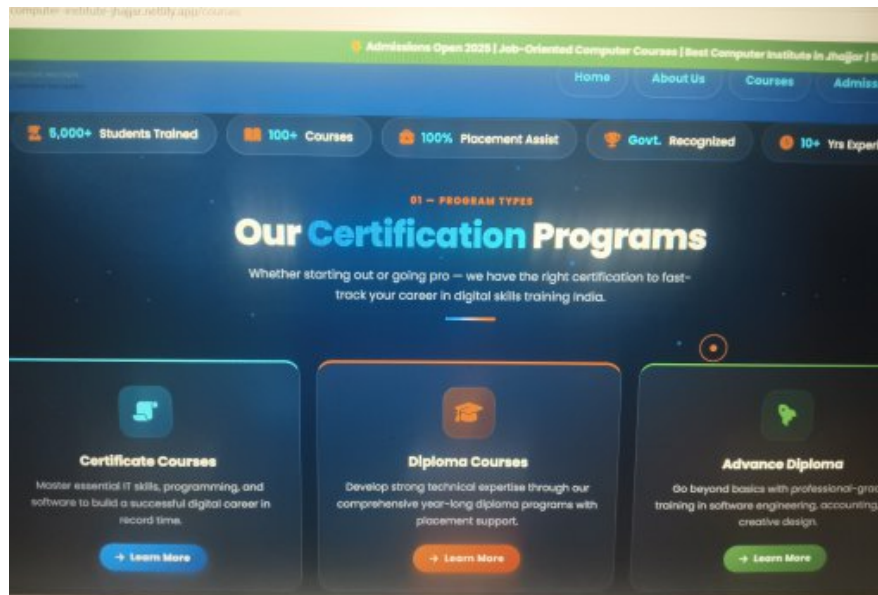


Figure 4.7: Courses page — Certificate, Diploma, and Advance Diploma programmes.

Figure 4.8 — About Us page (Welcome section): Going to “About Us” opens with a welcome message introducing Bharat Computer Institute and its mission of making quality computer education accessible to students from both urban and rural backgrounds. Alongside it is a short message from the Chairman, Ms. Parushi, and a “Our Training Excellence” panel showing completion rates for the institute's flagship courses, such as Computer Fundamentals and Tally with GST Training.

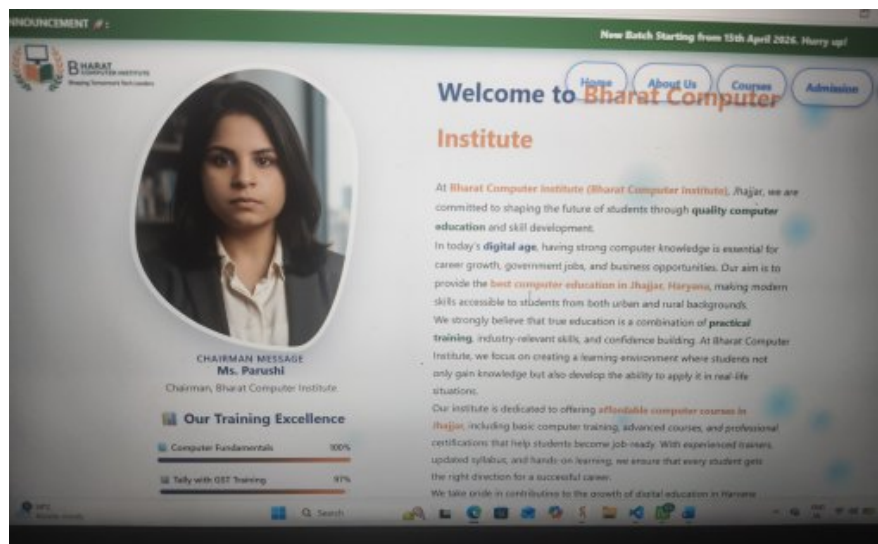


Figure 4.8: About Us page — welcome section and Chairman's message.

Figure 4.9 — About Us page (Institute overview): Scrolling further down the same page shows a photograph of the institute's training lab next to a description positioning Bharat Computer Institute as “Jhajjar's Most Trusted Computer Institute.” This section highlights the institute's AI-integrated teaching methods and its affiliation with E-Max India, which

gives its certifications national recognition.



Figure 4.9: About Us page — institute overview.

Figure 4.10 — About Us page (CEO's message): The final section of the About Us page carries a personal message from the CEO, Mr. Ritik Kumar, about the institute's belief that true learning happens when knowledge meets practical application. It closes the page on a direct, personal note before the visitor reaches the site footer.

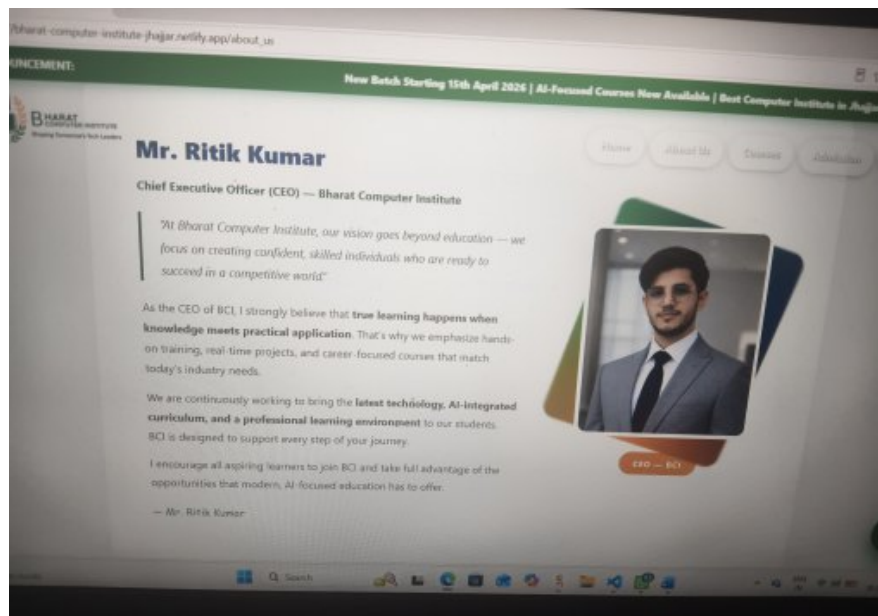


Figure 4.10: About Us page — CEO's message.

Note: the source-code snippets above (Figures 4.1–4.5) are representative sample implementations of the features described. Replace them with screenshots of the actual project source code before submitting.

4.9 Testing and Verification

The website was tested across both desktop and mobile screen sizes to confirm responsive behaviour. The contact form was tested with valid and invalid inputs to confirm that client-side validation correctly blocked incomplete submissions, and that valid submissions were correctly stored in the MySQL database via the Node.js API. The live Netlify deployment was checked to ensure it matched local testing behaviour.

4.10 Expected Outcomes

A working, responsive business/portfolio website built with HTML, CSS, and JavaScript.

A functioning Node.js/Express backend that accepts and processes form submissions.

A MySQL database correctly storing form data submitted through the website.

Supporting Python scripts used for testing and data handling during development.

A live, publicly accessible deployment of the website on Netlify.

Area	Key Skills Learned
Front-End	Responsive layout design, semantic HTML, CSS Flexbox/Grid, JavaScript DOM handling and form validation
Back-End	Building REST-style API routes with Node.js/Express, connecting to and querying a MySQL database
Tools & Deployment	Git/GitHub workflow, using Python for supporting scripts, deploying and configuring a live site on Netlify

Table 4.3: Skills learned summary.

4.11 Personal Learning Outcomes

This internship strengthened my practical understanding of full end-to-end web development — connecting a front end built with HTML, CSS, and JavaScript to a real backend and database, rather than working with static pages alone. I gained hands-on experience with Node.js and Express for server-side logic, MySQL for data storage, and Python for supporting scripts, and I learned how to take a completed project from local development through to a live deployment on Netlify.

4.12 Certificate of Completion

The Certificate of Internship (Web Development, 120 Hours, 90% Marks, A+ Grade) issued by the E-Max Council for Vocational Training and Education, through Bharat Computer Institute, is reproduced at the front of this report (see page iv).

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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This report was prepared as part of the Web Development internship at Bharat Computer Institute,
Jhajjar,
submitted in partial fulfilment of the B.Tech (CSE) programme at IILM University, Greater Noida.